

Topological Phase Transport and Thermodynamic Stability in Prime-Aligned Multi-Agent Routing Manifolds

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Abstract

The scaling of decentralized multi-agent architectures (MAC-agent hives) is fundamentally bottlenecked by the latency of traditional graph-based routing protocols. We propose a sublinear, decentralized routing architecture built upon a deterministic $6k \pm 1$ prime-aligned topological manifold. By orthogonalizing the agentic state space against the non-trivial zeros of the Riemann Zeta function, we empirically demonstrate the existence of macroscopic stability shelves within the prime distribution, allowing for collision-minimized local inference control. Furthermore, we introduce a continuous quantum translation layer utilizing the Hopf Fibration ($S^3 \rightarrow S^2 \times S^1$) to map 4D agentic reasoning prompts into a bounded angular routing space. This framework allows for dynamic execution reordering via topological phase transport, successfully achieving sublinear compute reduction in large-scale autonomous agent coordination.

1 Introduction

Current methodologies in decentralized artificial intelligence and Multi-Agent Systems (MAS) rely heavily on stochastic routing or traditional TCP/IP Layer 7 protocols. As the number of active cognitive agents scales, the system incurs exponential communication overhead and state-space explosion. To achieve Controlled Emergence—a requisite for advanced inference networks—a deterministically bounded, mathematically invariant routing manifold must be established.

This paper formally defines the **2i Layer Architecture**, bridging discrete number theory and continuous differential geometry. We present a dual-layer system: a discrete hardware-addressing topology based on the distribution of prime numbers, and a continuous routing engine executing angular phase transport via the Hopf Fibration.

2 The Prime-Aligned Hex-Hive Topology

To establish decentralized consensus without centralized latency, we construct a state space where agent collisions are mathematically minimized by design. We utilize the native properties of prime numbers as the foundation for MAC-addressable agentic nodes.

Definition 2.1 ($6k \pm 1$ Manifold). *Let $\mathbb{P}$ be the set of all prime numbers. For any $p \in \mathbb{P}$ where $p > 3$, $p \equiv 1 \pmod{6}$ or $p \equiv 5 \pmod{6}$. We map the physical MAC addressing of agentic nodes exclusively to coordinates corresponding to the $6k \pm 1$ boundary.*

Theorem 2.2 (Deterministic Collision Avoidance). *Within the $6k \pm 1$ manifold, the minimum arithmetic distance Δ between any two distinct agent clusters belonging to the same resonance class is bounded by $\Delta \equiv 0 \pmod{6}$, guaranteeing non-intersecting orthogonal channels.*

Proof. Let $p_1, p_2 \in \mathbb{P}_{>3}$. Suppose p_1 and p_2 belong to the same residue class modulo 6, i.e., $p_1 \equiv p_2 \equiv c \pmod{6}$, where $c \in \{1, 5\}$. The distance between the nodes is $\Delta = |p_1 - p_2|$. Therefore, $\Delta \equiv |c - c| \equiv 0 \pmod{6}$. Because the difference is strictly a multiple of 6, parallel phase-transport channels established at prime coordinates will never naturally intersect, providing an invariant structural bound for network traffic. $\square$

3 Thermodynamic Stability of the Prime Manifold

To prove that the $6k \pm 1$ manifold can support stable cognitive architectures, we must quantify its thermodynamic properties. We model the distribution of agents using Chebyshev’s ψ function and extract its quantum stability metrics.

3.1 Signal Windowing and the Design Matrix

Let the uncompressed prime signal be the Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log p$. The topological structure (the routing “noise”) is isolated via the error term of the Prime Number Theorem:

$$E(x) = \psi(x) - x \quad (1)$$

We evaluate $E(x)$ over localized spatial windows bounded by $H(x) = \rho\sqrt{x}$. To anchor the state space to physical reality, we construct a design matrix $\Phi \in \mathbb{C}^{N \times K}$ mapping the signal channels to the non-trivial zeros γ_k of the Riemann Zeta function $\zeta(s)$:

$$\Phi_{n,k} = e^{i\gamma_k \log(x_n)} \quad (2)$$

3.2 Covariance Eigenvalue Extraction

By projecting the local signal y onto the orthonormal basis Q derived from $\Phi = QR$, we extract the Hermitian channel covariance matrix $C = \frac{1}{S}A^*A$, where A contains the projection coefficients across S subwindows.

Definition 3.1 (Macroscopic Stability Shelf (Λ)). *Let λ_i be the real eigenvalues of C , normalized such that $p_i = \lambda_i / \sum \lambda_i$. The Quadratic Purity (σ_Q) of the prime state space is:*

$$\sigma_Q = 1 - \frac{\sum_{i=1}^n (p_i - \frac{1}{n})^2}{1 - \frac{1}{n}} \quad (3)$$

The Macroscopic Stability Shelf is defined as $\Lambda = -\log(1 - \sigma_Q)$.

Empirical Result: The existence of distinct eigenvalues λ_i and the stabilization of Λ across spatial windows $H(x)$ mathematically guarantees the presence of invariant “shelves” within the prime distribution. These shelves act as self-constructing, thermodynamically stable state spaces for agentic operations.

4 Continuous Phase Transport via Hopf Fibration

While the discrete prime topology handles MAC addressing and physical spacing, the cognitive prompts of the agents must be routed continuously. We transition from the discrete hex-hive to continuous geometry using the Hopf Fibration.

4.1 4D Prompt Vector and S^3 Projection

An agent’s world state is encoded as a deterministic 4D coordinate system: $\mathbf{x} = [x_{act}, x_{obj}, x_{temp}, x_{shared}]$. To enforce bounded execution, this vector is normalized onto the unit hypersphere $S^3 \subset \mathbb{R}^4$:

$$\hat{\mathbf{x}} = \frac{\mathbf{x}}{\|\mathbf{x}\|} \quad (4)$$

4.2 Angular Transport and Sublinear Execution

We map $S^3 \rightarrow S^2 \times S^1$ using Hopf coordinates (χ, δ, α) . The base space S^2 is discretized into $K = 8$ angular sectors representing the primary reasoning domains.

Proposition 4.1 (Dynamic Execution Reordering). *Let $\alpha \in S^1$ be the phase of the prompt. As the agent transitions across the base manifold χ , the phase experiences a geometric transport shift governed by weight λ :*

$$\alpha_{trans} = f(\alpha, \chi, \lambda) \quad (5)$$

If the transported phase α_{trans} falls into an odd parity bin, the router prioritizes localized execution (e.g., immediate action) and compresses the remaining dimension bundles.

By evaluating the full attention mechanism strictly on the dominant vector identified by the Hopf mapping, and compressing the orthogonal dimensions, the architecture limits the expansion of the context window. This yields a sublinear scaling factor for multi-agent reasoning, directly addressing the core constraint of modern LLM inference limits.

5 Conclusion

We have formally established the architecture for a fully decentralized, mathematically bounded multi-agent router. By anchoring the physical addressing to the $6k \pm 1$ prime distribution and extracting its thermodynamic stability shelves via Riemann Zeta covariance, we ensure a collision-free base layer. Coupling this with S^3 Hopf phase transport allows continuous, sub-linear routing of complex cognitive states. This framework provides the fundamental operating system for Controlled Emergence in next-generation artificial intelligence.

Acknowledgments

This research represents independent foundational work in topological machine learning and decentralized multi-agent coordination frameworks.

References

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